

Vivek Devulapalli, Ph.D.

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🌐 <https://www.linkedin.com/in/vivek-devulapalli-stem/>

🔍 <https://scholar.google.com/citations?user=TE3lrxoAAAAJ&hl=en>

🌐 <https://vivekdevulapalli07.github.io/website/>



Professional summary

I am an early-career postdoctoral researcher specializing in advanced electron microscopy with a materials science background. I employ aberration-corrected STEM to link atomic structure and composition to material properties across multiple length scales. My expertise includes multidimensional imaging, analytical techniques, STEM simulations, in-situ nanomechanics, and correlative approaches for comprehensive material characterization.

My PhD work on grain boundary engineering in titanium, including the discovery of segregation-induced phase transitions, culminated in a Science publication (2024). As a postdoc at Max-Planck (MPIE), I studied hydrogen embrittlement in particle-reinforced high-entropy alloys by combining advanced characterization with mechanical testing. At EMPA, I use in-situ S/TEM to investigate deformation from single crystals to multilayered thin films. I am passionate about teaching and actively create interactive content for scientific dissemination, freely shared on GitHub. I also have experience in proposal writing and mentoring students, which prepares me well for upcoming academic challenges. I strongly support open-source Python-based data analysis and rigorous research data management with electronic lab notebooks.

Scientific career

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| 2026 – Present | <div>📌 Postdoctoral researcher, EMPA, Thun, Switzerland</div> <div>EMPA Young Scientist Fellowship (CHF 270,000): First major grant as PI, <i>Mapping nanoscale deformation mechanics in amorphous oxide multilayer coatings (MOSAIC)</i></div> <div>Currently preparing an Innosuisse project proposal with Inspire AG and Dyconex AG; initiated this collaboration by independently connecting both companies with Empa.</div> <div>PI: Prof. Johann Michler, Dr. Xavier Maeder</div> |
| 2024 – 2026 | <div>📌 Postdoctoral researcher, EMPA, Thun, Switzerland</div> <div>Project: <i>In-situ TEM tensile testing of crystalline-amorphous nanolayered materials</i></div> <div>SNSF Ambizione proposal: <i>In situ nanomechanics and phase transformations at interfaces (INSPIRE)</i></div> <div>MecaNano Short-Term Scientific Mission Grant (EUR 3,000): Research visit with Dr. Marc Legros, CEMES, Toulouse, France (3 weeks, February 2026)</div> <div>PI: Prof. Johann Michler, Dr. Xavier Maeder</div> |
| 2022 – 2024 | <div>📌 Postdoctoral researcher, Max-Planck Institut für Eisenforschung, Düsseldorf, Germany</div> <div>DFG Project: <i>Mechanical properties and hydrogen tolerance of particle-reinforced CCA produced by additive manufacturing</i></div> <div>PI: Prof. Gerhard Dehm, Prof. Eric Jägle</div> |
| 2018 – 2022 | <div>📌 PhD, Max-Planck Institut für Eisenforschung, Düsseldorf, Germany</div> <div>Advanced transmission electron microscopy group, Department of structure and nano-/ micromechanics of materials</div> <div>Thesis title: <i>Microstructure and grain boundary evolution in titanium thin films.</i></div> <div>Supervisor: Prof. Gerhard Dehm, Group leader: Dr. Christian Liebscher</div> |
| 2016 – 2018 | <div>📌 M.Tech., Indian Institute of Technology, Madras, India</div> <div>Metallurgical and materials engineering department</div> <div>Thesis title: <i>Correlative microscopy of Magnesium - rare-earth alloys.</i></div> <div>Thesis work performed as a DAAD exchange program at Materials Chemistry, RWTH Aachen, Germany with Prof. Jochen Schneider</div> <div>Supervisors: Prof. B S Murty, Prof. K G Pradeep</div> |
| 2012 – 2016 | <div>📌 B.Tech., National Institute of Technology, Raipur, India in Metallurgical engineering</div> <div>CGPA: 9.22/10 (Honours)</div> |

Research Publications

First Author Publications

- 1 **Devulapalli, V.**, Klimashin, F. F., Baertschi, M., Waldner, S., Schwyn Thoeny, S., Michler, J., & Maeder, X. (2026). Interface-mediated softening and deformation mechanics in amorphous/amorphous nanolaminates. *Scripta Materialia*, 255, 117118. <https://doi.org/10.1016/j.scriptamat.2025.117118>
- 2 **Devulapalli, V.**, Prabhakar, M., Schulz, F., Maeder, X., Jaegle, E. A., Duarte, J., & Dehm, G. (2026). Hydrogen induced softening and deformation mechanisms in nanoparticle reinforced CoCrFeNi [Under preparation].
- 3 **Devulapalli, V.**, Vermeij, T., Sharma, A., Michler, J., & Maeder, X. (2026). Enhanced toughness in cu-al nanolaminate films with amorphous al₂O₃ interlayers: Insights from in situ tem tensile testing [Under preparation]. <https://doi.org/10.5281/zenodo.17523627>
- 4 **Devulapalli, V.**, Schulz, F., Barreto, E. S., Ellendt, N., Jaegle, E. A., & Dehm, G. (2024). Laser powder bed fusion synthesis of nanoparticle reinforced cocrfeni. *Additive Manufacturing*, 91, 104338. <https://doi.org/10.1016/j.addma.2024.104338>
- 5 **Devulapalli, V.**, Chen, E., Brink, T., Frolov, T., & Liebscher, C. (2024). Segregation induced grain boundary phase transition in titanium. *Science*, 386, 420–424. <https://doi.org/10.1126/science.adq4147>
- 6 **Devulapalli, V.**, Hans, M., Sukumar, P. T., Schneider, J. M., Dehm, G., & Liebscher, C. (2022). Microstructure, grain boundary evolution and anisotropic fe segregation in (0001) textured ti thin films. *Acta Materialia*, 238, 118180. <https://doi.org/10.1016/j.actamat.2022.118180>
- 7 **Devulapalli, V.**, Bishara, H., Ghidelli, M., Dehm, G., & Liebscher, C. (2021). Influence of substrates and e-beam evaporation parameters on the microstructure of nanocrystalline and epitaxially grown ti thin films. *Applied Surface Science*, 562, 150194. <https://doi.org/10.1016/j.apsusc.2021.150194>

Co-authored Publications

- 1 Sahu, S., **Devulapalli, V.**, Saood, S., Dehm, G., & Best, J. P. (2026). Impact of focused ion beam-induced damage on micromechanical properties in hematite. *Scripta Materialia*, 257, 117000. <https://doi.org/10.1016/j.scriptamat.2025.117000>
- 2 Kroeker, P., Vermij, T., **Devulapalli, V.**, Michler, J., & Maeder, X. (2026). Secondary twinning and twin-twin interaction mechanisms of {11 $\bar{2}$ 1} deformation twins in rhenium [Accepted in Acta Materialia]. *Acta Materialia*.
- 3 Lyrio, M., Oliveira, H., Sandim, M., **Devulapalli, V.**, & Sandim, H. (2025). Effect of the scanning strategy on texture of grain-oriented electrical steel (fe-4wt.% si) processed via laser powder-bed fusion and subsequent thermomechanical processing. *Materials Characterization*, 221, 114789. <https://doi.org/10.1016/j.matchar.2025.114789>
- 4 Chatzopoulou, P., Hilliard, D., Vasileiadis, I. G., Florini, N., **Devulapalli, V.**, Liebscher, C. H., Lymperakis, L., Komninou, P., Kehagias, T., & Dimakis, E. (2025). Elastic limit and relaxation of gaas/in(al,ga)as core/shell nanowires for near-infrared applications [Focus on Nanowires 2023-24]. *Nanotechnology*, 36(9), 095703. <https://doi.org/10.1088/1361-6528/ad9d4a>
- 5 Baertschi, M., Waldner, S., Schwyn Thoeny, S., Maeder, X., Steger, F., Frei, T., & **Devulapalli, V.** (2025). Uv coatings using ta₂O₅-sio₂ quantized nanolaminates. *Journal of the European Optical Society-Rapid Publications*, 21(1), 24. <https://doi.org/10.1051/jeos/2025021>
- 6 Byloff, J., **Devulapalli, V.**, Casari, D., Edwards, T. E. J., Trost, C. O. W., Cordill, M. J., Husain, S. A., Renault, P.-O., Faurie, D., & Putz, B. (2025). From mechanics to electronics: Influence of ald interlayers on the multiaxial electro-mechanical behavior of metal-oxide bilayers. *Advanced Functional Materials*, 36(27), e26343. <https://doi.org/10.1002/adfm.202526343>
- 7 Byloff, J., Trost, C. O. W., **Devulapalli, V.**, Altaf Husain, S., Faurie, D., Renault, P.-O., Edwards, T. E. J., Cordill, M. J., Casari, D., & Putz, B. (2025). Atomic layer-deposited interlayers for robust metal-polymer interfaces. *ACS Applied Materials & Interfaces*. <https://doi.org/10.1021/acsami.5c05156>
- 8 Thoeny, S. S., Baertschi, M., Waldner, S., Frei, T., & **Devulapalli, V.** (2025). Form birefringence in magnetron sputter deposited nanolaminates of ta₂O₅-sio₂. *Advances in Optical Thin Films IX*, 14107, 198.

- 9 Senol, S., Li, G., **Devulapalli, V.**, Brodu, E., & Vanmeensel, K. (2024). High strength and fatigue performance achieved for l-pbf processed hybrid particle reinforced al-cu-mg composite. *Composites Part B: Engineering*, 285, 111736.
<https://doi.org/10.1016/j.compositesb.2024.111736>
- 10 Dhekne, P. P., Vermeij, T., **Devulapalli, V.**, Jadhav, S. D., Hoefnagels, J. P., Geers, M. G., & Vanmeensel, K. (2023). Micro-mechanical deformation behavior of heat-treated laser powder bed fusion processed ti-6al-4v. *Scripta Materialia*, 233, 115505. <https://doi.org/10.1016/j.scriptamat.2023.115505>
- 11 Gaertner, E., Witte, A., Peter, N. J., **Devulapalli, V.**, Ellendt, N., Dehm, G., Jaegle, E. A., Uhlenwinkel, V., & Maedler, L. (2023). Melt pool signatures of tin nanoparticle dry-coated $Co_{25}Cr_{25}Fe_{25}Ni_{25}$ metal powder in laser-powder-bed-fusion. *Materials & Design*, 226, 111626. <https://doi.org/10.1016/j.matdes.2023.111626>
- 12 Vasileiadis, I. G., Lymperakis, L., Adikimenakis, A., Gkotlinakos, A., **Devulapalli, V.**, Liebscher, C. H., Androulidaki, M., Huebner, R., Karakostas, T., Georgakilas, A., Komninou, P., Dimakis, E., & Dimitrakopoulos, G. P. (2021). Substitutional synthesis of sub-nanometer InGaN/GaN quantum wells with high indium content. *Scientific Reports*, 11(1), 20606. <https://doi.org/10.1038/s41598-021-99989-0>
- 13 Wu, G., Liu, C., Brognara, A., Ghidelli, M., Bao, Y., Liu, S., Wu, X., Xia, W., Zhao, H., Rao, J., Ponge, D., **Devulapalli, V.**, Lu, W., Dehm, G., Raabe, D., & Li, Z. (2021). Symbiotic crystal-glass alloys via dynamic chemical partitioning. *Materials Today*, 51, 6–14.

Skills

Instruments with 7+ years of hands-on experience

■ Thin film deposition, Material characterization on all length scales - XRD, SEM, EDS, EBSD, FIB, TEM, Multi-dimensional STEM techniques (EELS, ACOM, 4DSTEM), APT.

Programming and softwares

■ Python - multidimensional microscopy data analysis and image processing (py4Dstem, hyperspy), image simulation (abTEM, prismatic), image processing (numpy, scikitimage), machine learning and image classification/segmentation (PyTorch), data visualization (mpl), L^AT_EX, Crystallographic analysis (MTEX), Ovito, Inkscape, Adobe Premier pro .

Conferences

Talk

- ICMCTF 2027, San Diego, CA, USA (April 25–30, 2027)
Invited talk: Direct observation of toughening mechanisms in Cu(Al) nanolaminate thin films with amorphous Al₂O₃ interlayers via in situ TEM tensile testing.
- TMS Annual Meeting & Exhibition, Orlando, FL, USA (March 14–18, 2027)
 Microstructural control of hydrogen transport and deformation mechanisms in LPBF-fabricated nanoparticle-reinforced CoCrFeNi composites.
- TMS Annual Meeting & Exhibition, Orlando, FL, USA (March 14–18, 2027)
 Strength-toughness synergy in crystalline-amorphous nanolaminates via ALD interlayer engineering revealed by in situ TEM imaging.
- TMS Annual Meeting & Exhibition, San Diego, CA, USA (March 2026)
 Interface-Mediated Deformation Transition and Quantum Confinement Effects in Ta₂O₅/SiO₂ Optical Nanolaminates.
 Session: Interface-Driven Plasticity and Fracture in Nanostructured and Layered Materials.
- TMS Annual Meeting & Exhibition, San Diego, CA, USA (March 2026)
 Hydrogen-Enhanced Localized Plasticity in Nanoparticle-Reinforced CoCrFeNi Medium-Entropy Alloys.
 Session: Alloy Synthesis and Processing.
- TMS Annual Meeting & Exhibition, San Diego, CA, USA (March 2026)
 In-Situ TEM Investigation of Fracture Growth in Novel Nanolaminate Thin Films.
 Session: Advanced Characterization Techniques for Quantifying and Modeling Deformation - Electron Microscopy.

Conferences (continued)

- EMPA Department o2o Kick-off Meeting (January 2026)
In-Situ Nanomechanics and Electron Diffraction in TEM: Revealing Deformation Mechanisms in Nanolaminate Structures.
- In situ Electron Microscopy Workshop, Toulouse, France (January 2026)
Advanced in situ TEM techniques for nanomaterial characterization.
- EMPA Board of Governors Department 206 Meeting (September 2025)
Probing deformation mechanics at micro-/nanoscale.
- FEMS-Euromat, Granada, Spain (2025)
Hydrogen induced softening and deformation mechanisms in nanoparticle reinforced CoCrFeNi.
- FEMS-Euromat, Granada, Spain (2025)
In situ TEM investigation of fracture growth in novel nanolaminate thin films.
- MecaNano - General meeting, Krakow, Poland (2025)
Fracture behaviour in Cu-Al multilayer thin films with amorphous AlO interlayers: Insights from in-situ TEM tensile testing.
- International Conference on Correlative Electron Microscopy, Brno, Czech Republic (2025)
Enhancing Thin Film Toughness: The Role of Amorphous Interlayers.
- FEMS - Euromat, Frankfurt, Germany (2023)
Additively manufactured particle reinforced-CoCrFeNi: Microstructure, mechanical behaviour and hydrogen response.
- MSE Darmstadt (Virtual), Darmstadt, Germany (2020)
Unravelling grain boundary structures in Ti thin films using aberration-corrected transmission electron microscopy.
- Poster ■ International Microscopy Conference (IMC20), South-Korea (2023)
Experimental observation of segregation induced grain boundary phase transition in titanium
- Microscopy Conference 2021, Vienna, Austria (Virtual) (2021)
Unravelling the atomic structure and segregation of Σ_{13} [0001] tilt grain boundaries in titanium by advanced STEM.
- IAMNano, Düsseldorf, Germany (2019)
From epitaxially grown thin films to grain boundary analysis in Cu and Ti.

Positions of Responsibility

Supervision and teaching experience

1. **Guest lecturer for Master's students at University of Bern in Precision Engineering: Delivered 3 lectures on "Advances in Transmission Electron Microscopy", 2024-2025.**
2. Supervised a Cambridge University summer intern for 8 weeks, 2023.
3. Peer-reviewed a full-length journal article for *Materialia*, 2023.
4. Delivered a lecture on 'grain boundaries' as part of a seminar series at MPIE, 2021.
5. Served as a teaching assistant under Prof. Ravi Shankar Kottada, IIT - Madras, 2016.

Leadership & Organization

1. Member of European Microscopy Society (EMS) and Deutsche Gesellschaft für Elektronenmikroskopie e.V. (DGE), 2019-present.
2. Internal PhD representative, 2019-2022.
3. Member of the Max-Planck Institute für Eisenforschung (MPIE) sustainability team, 2021-2023.
4. Organized and participated in a teaching seminars and career planning seminar, 2021-2022.
5. Managed MPIE's Twitter account, 2020-2023.
6. Coordinated department weekly-seminars, 2022-23.

7. Managed a high-vacuum furnace, 2021-2023.

Science communication

1. Actively developing open-source educational resources including interactive web-based tools for materials science concepts and Jupyter notebooks for electron microscopy data analysis, available on GitHub and personal website.  <https://vivekdevulapalli07.github.io/website/>  <https://github.com/vivekdevulapalli07>
2. Represented MPIE at the Bright World of Metals science slam event, 2023.
3. Organized and spoke at "Pint of Science," Düsseldorf, 2021 and 2023.
4. Represented MPIE at "Nacht-der-Wissenschaft (Science Night)," Düsseldorf, 2022.
5. Attended the **Alexander-von-Humboldt scholarship-funded** Science Communication Summer School in Berlin, 2021, resulting in a published *Declaration on the Future of Science Communication* presented to the German Parliament.
6. Editorial team member for the Max-Planck PhD-net scientific magazine, "Offspring," 2019-2020.

Awards and recognition

- **EMPA Young Scientist Fellowship**, CHF 270,000, 2026.
- **MecaNano Short-Term Scientific Mission (STSM) Grant**, EUR 3,000, 2026.
- Granted travel support by MecaNano cost action to give talk at Euromat 2025.
- Granted travel support by MecaNano cost action to give talk at MecaNano - General meeting, 2025.
- Granted travel support by European Microscopy Society for MC-2021, Vienna.
- **DAAD IIT-Masters scholarship** for a 7 month research stay in RWTH-Aachen, Germany, 2017.
- Scored **All India Rank – 76** in Graduate Aptitude Test in Engineering in Material Science, 2016.
- Recognitions in 2015–2016 include scoring **All India Rank - 4** in the Metallurgical Thermodynamics examination conducted by NPTEL, securing an ARCI Hyderabad summer internship through a first-place presentation at IIT - BHU, and being awarded **Eminent student of the branch** by the Humanities department at NIT – Raipur.

References

Prof. Dr. Gerhard Dehm

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Department of structure and nano-/micromechanics of materials,
MPIE, Düsseldorf
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Prof. Dr. Christian Liebscher

Professor of Advanced Transmission Electron Microscopy at the Faculty of Physics and Astronomy and the Research Center Future Energy Materials and Systems (RC FEMS),
Ruhr University Bochum
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Prof. Dr. Johann Michler

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